window.*input_file = new TextDecoder('utf-8').decode( Uint8Array.from(atob('LS0tCnRpdGxlOiAiQWJzdHJhY3Rpb24gJiBDb250cmFzdCIKZm9yb
function(c) { return c.charCodeAt(0); }) ); window.*input_filename = '/Users/joseffruehwald/Documents/courses/Lin300-2026.github.io/course-materials/lecture-notes/15_abstraction/index.qmd'; </script>

# Abstraction & Contrast

Dr. Josef Fruehwald

## From Phonetics to Phonology

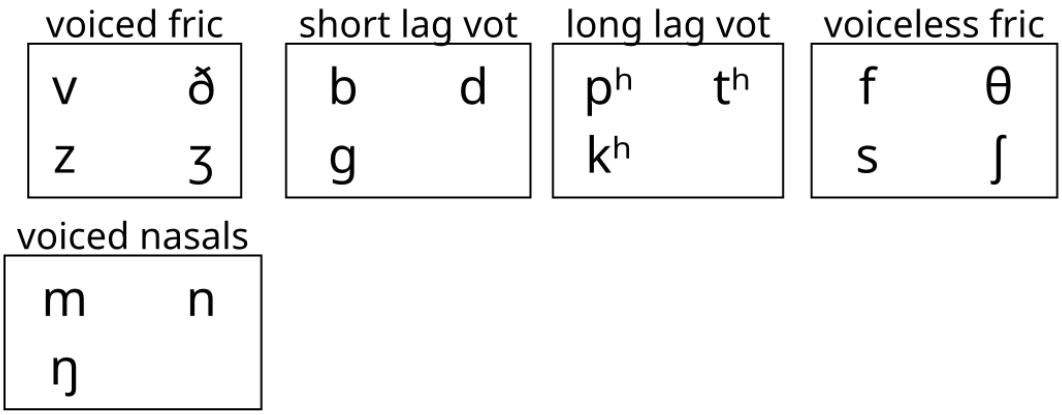
```
Loading required package: sysfonts
```

```
Loading required package: showtextdb
```

There is no definition of the difference between Phonetics and Phonology that won't make *someone* angry.

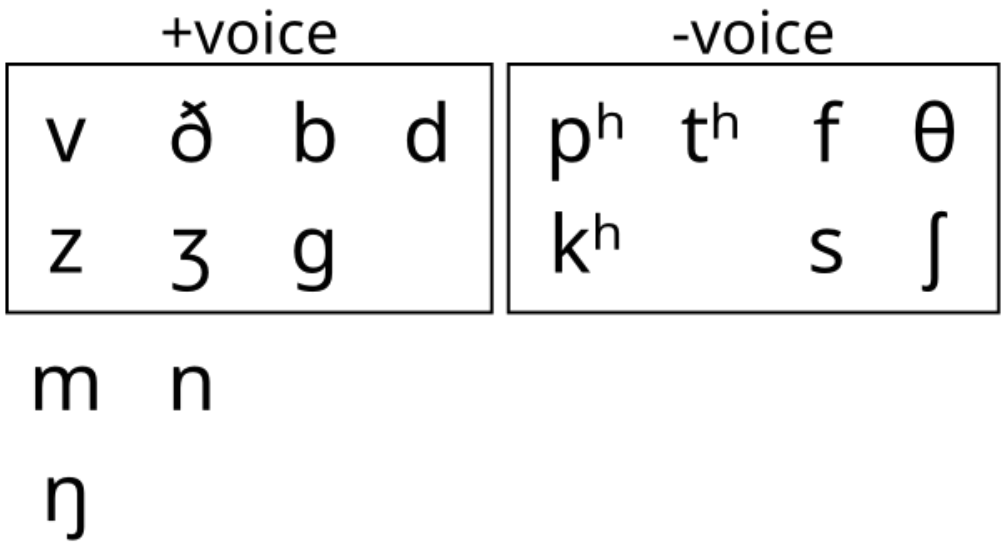
## Phonetics

English Consonants




## Phonology

English Consonants




## What do you know?

Linguistics is all about answering the question “What is your knowledge of your language?”

## Aside: Knowledge

## Knowledge of sounds

long-lag VOT



short-lag VOT



Knowledge of Sounds

How would you transcribe your pronunciation of a novel word like:

# “kowspie”

Knowledge of Sounds

| | | |
|---|---|---|
| | $k^h$ | k |
| $p^h$ | $k^hosp^hi$ | $kosp^hi$ |

p kʰospi kospi

## Knowledge of Sounds

Your preference (I'm guessing!) for [kʰospi], demonstrates a kind of knowledge about the sounds of language beyond just the fact that there are long-lag and short-lag VOT segments.

## Aside: Wugs

## Knowledge of Sounds

| | | | |
|---|---|---|---|
| *play* | ˈ**p**ʰleɪ | dɪ.ˈs**p**leɪ | *display* |
| *card* | ˈ**k**ʰɑɪd | dɪ.ˈs**k**ɑɪd | *discard* |

## Knowledge of Sounds

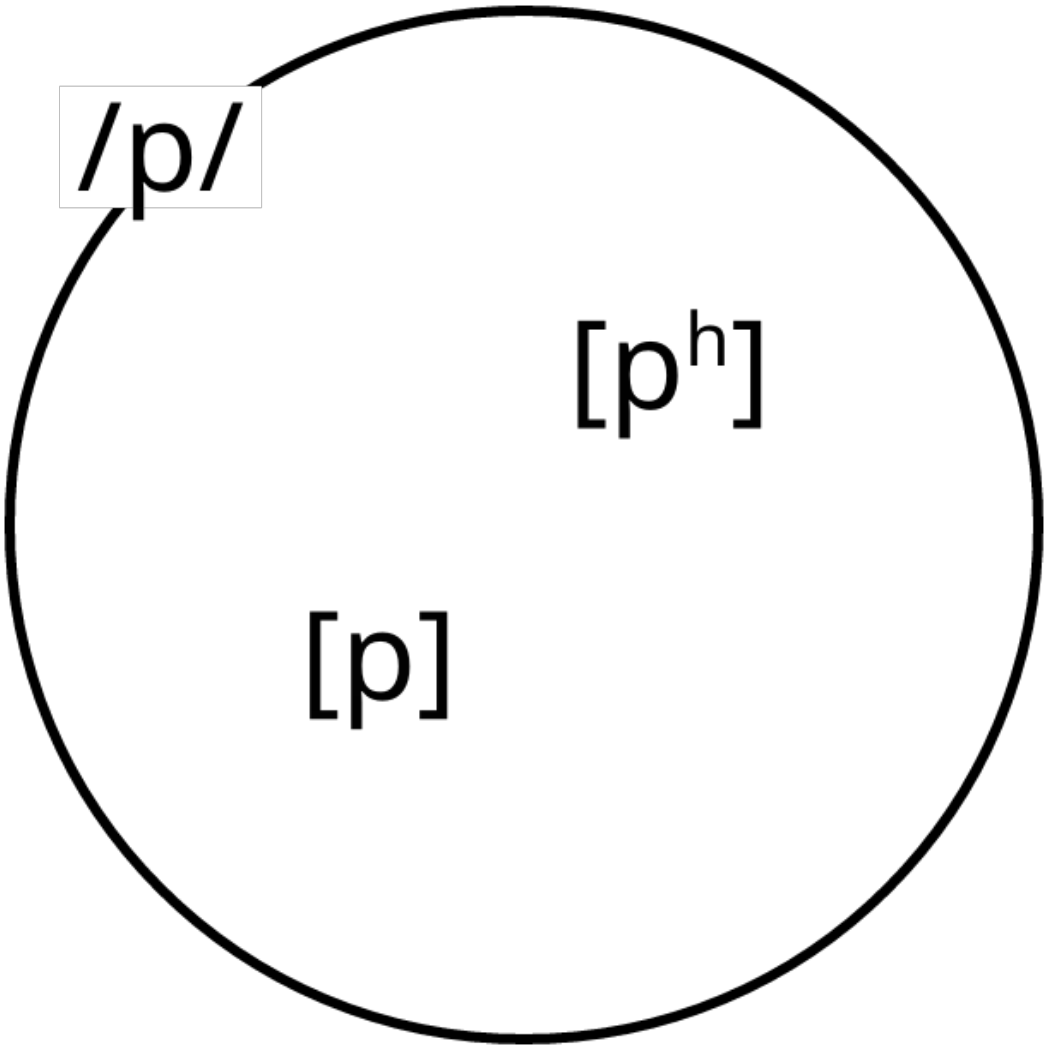
| | | |
|---|---|---|
| *long* | *longing* | *longer* |
| lɔŋ | lɔŋɪŋ | lɔŋgɻ |

## Knowledge of Sounds

Upshot: Abstraction is necessary

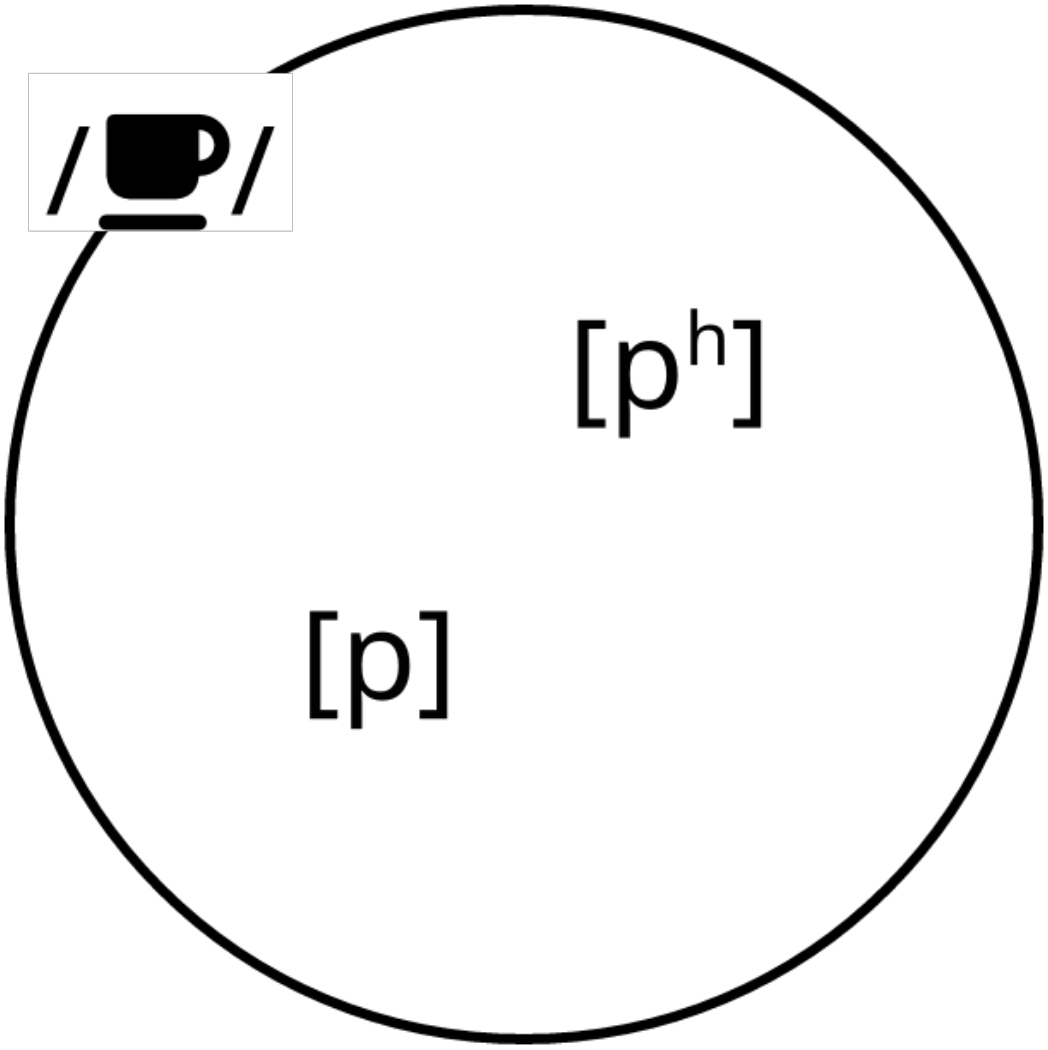
## "Phoneme"

A label for a set of sounds that all count as basically the same.




The phoneme /p/ has at least two *allophones*, [p] and [pʰ]

“Phoneme”




We use “/p/” as convenient label for the phoneme, but the phoneme itself is not a sound.

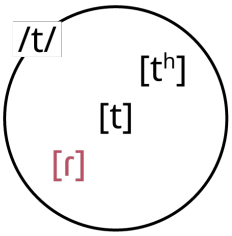
We could have used “” as the label for the phoneme, but it would be more confusing to read.

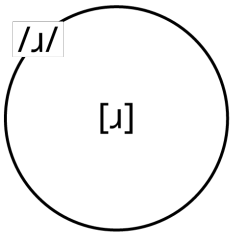
## Allophones

- Not every phoneme has the same number of allophones.
- There isn’t a universal set of relationships between allophones and phonemes.

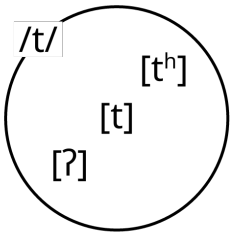
## Allophones

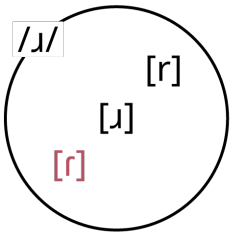
American English




Scottish English



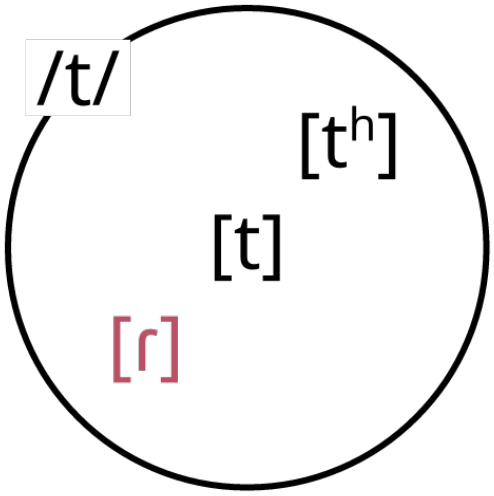



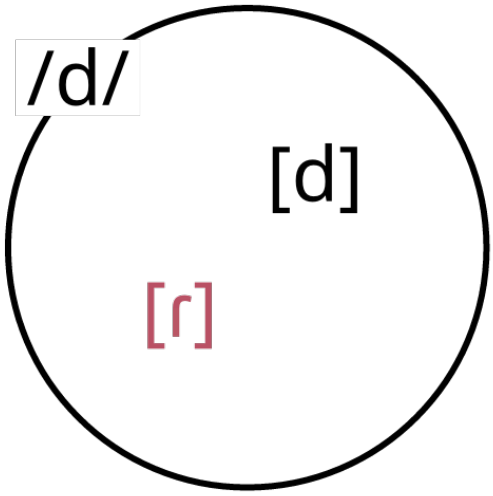

## Allophones

There isn't always a 1-to-1 relationship between phones and phonemes.

## Allophones

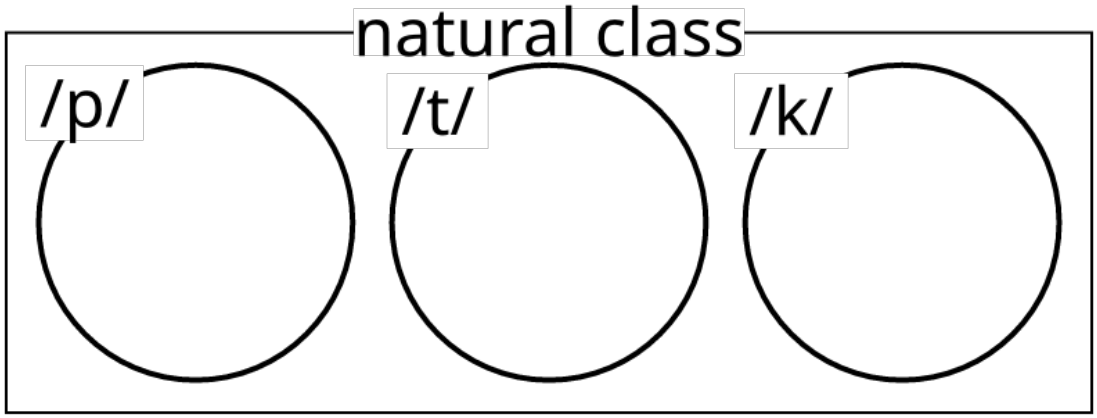
American English




## "Natural Classes"

Groups of phonemes that pattern together are "natural classes"




## Allophones & Natural Classes

Which sounds are allophones of which phonemes, and which phonemes form natural classes, are all language specific.

## Phonemic Contrast

If two sounds are *not* basically the same, then they are *contrastive*, and are allophones of different phonemes.

| *ether* | *either* |
|---|---|
| [iθɻ] | [iðɻ] |

## Phonemic Contrast

### Minimal Pairs

Two words that are identical except for the one phoneme in question

# d

?

# ð

?

## Allophonic Alternation

### Complementary Distribution

If two sounds never appear in the same contexts, they are in a complementary distribution.

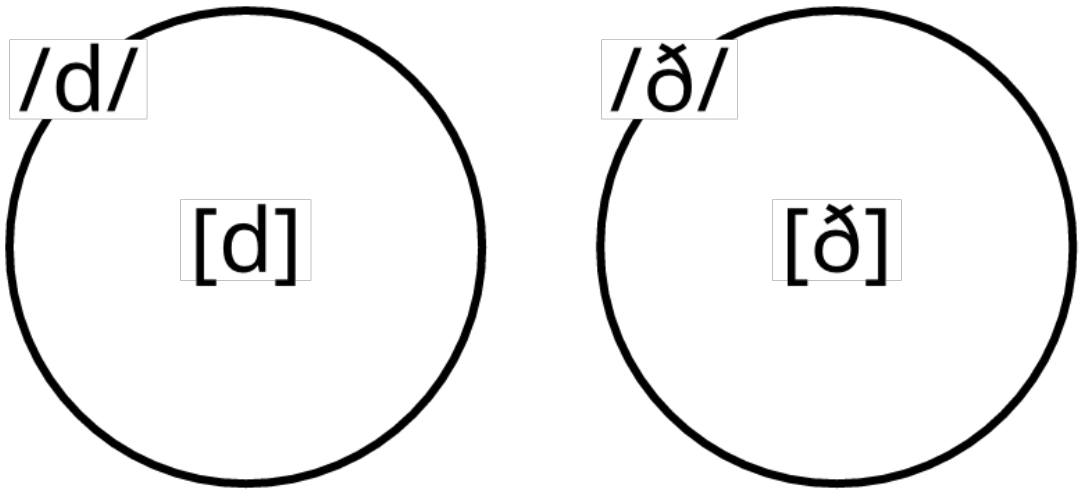
## Allophonic Alternation

Spanish:

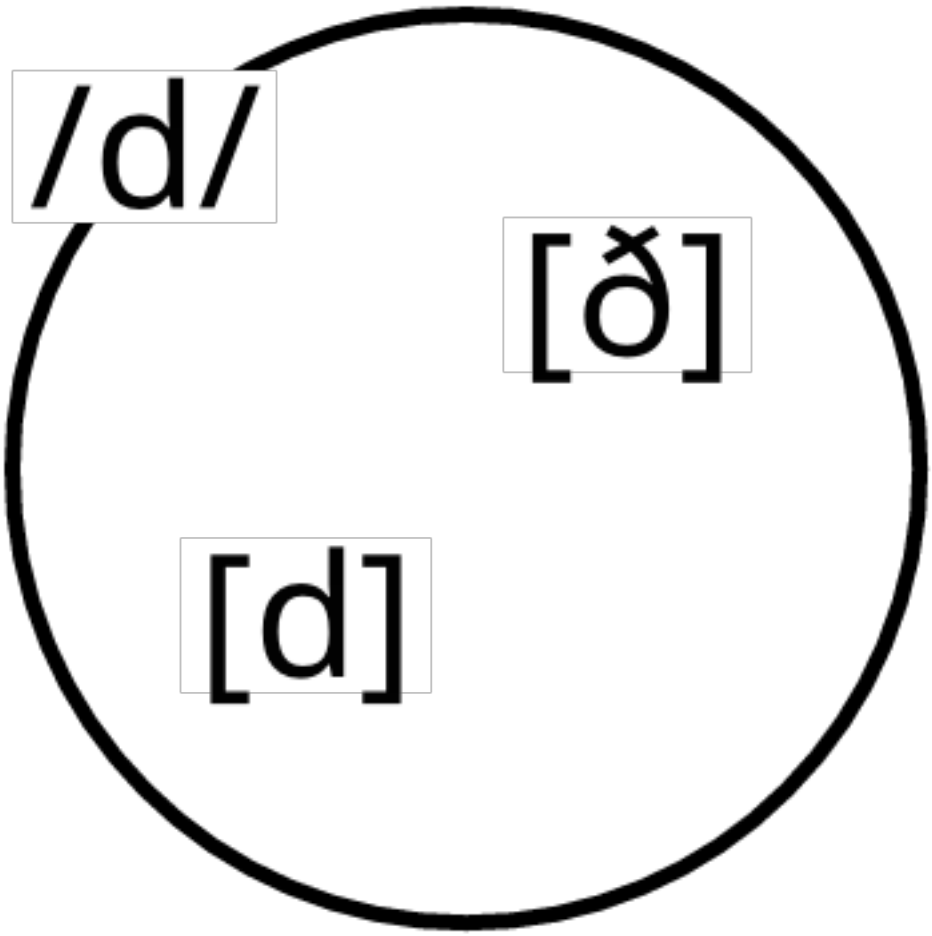
| | |
|---|---|
| [dama] | *lady* |
| [dulse] | *sweet* |
| [disfrutas] | *disguise* |
| [laðo] | *side* |
| [universiðað] | *university* |
| [granaða] | *Granada* |

## Phonological analysis

English:




Spanish:


/d/
[ð]
[d]